MIL-A-49273(CR)
26 January 1988

MILITARY SPECIFICATION

AMPLIFIER, POWER SUPPLY AM-7152/GRC-213 AND ADAPTER KIT, VEHICLE MK-2154/GRC-213

This specification is approved for use within the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the Amplifier, Power Supply AM-7152/GRC-213 and Adapter Kit, Vehicle MK-2154/GRC-213 which together are used for vehicular operation of the AN/PRC-104 Radio Set (Reference to AN/PRC-104 also includes AN/PRC-104A, AN/PRC-104(V) and AN/PRC-104B(V)). Performance specifications of the AN/PRC-104 are provided in specification MIL-R-29418 (MC). This equipment allows operation of the AN/PRC-104 in a moving wheeled or tracked vehicle, using the nominal 28 volt vehicular supply. The amplifier-power supply also provides operation of the AN/PRC-104 with a vehicular intercommunication system (AN/VIC-1(V)), a squelched one-watt audio drive for a loudspeaker LS-454, and antenna conditioning for a 15 foot whip antenna (base AB-652B, three each sections MS-116, and one each of sections MS-117 and MS-118).

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army Communications-Electronics Command and Fort Monmouth, ATTN: AMSEL-ED-TO, Fort Monmouth, New Jersey 07703-9990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC: N/A FSC 5820

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SPECIFICATIONS

FEDERAL

PPP-B-636	Boxes, Shipping, Fiberboard

MILITARY

MIL-C-22750	Coating, Epoxy-Polymide
MIL-C-46168	Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-C-55442	Cable Assemblies and Cord Assemblies, Packaging of
MIL-E-16400	Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore, General Specification for
MIL-E-55585	Electronic Equipment and Parts, Packaging of
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-L-12632	Loudspeaker, Permanent Magnet (Encased, 2 Watt, Fungus, Gunblast, and Immersion Resistant), General Specification for
MIL-P-116	Preservation, Methods of
MIL-P-11268	Parts, Materials and Processes in Electronic Equipment
MIL-P-23377	Primer Coating: Epoxymide, Chemical and Solvent Resistant
MIL-Q-9858	Quality Program Requirements
MIL-R-29418	Military Specification Radio Set: AN/PRC-104(V)
MIL-S-901	Shock Tests, High-Impact Shipboard Machinery, Equipment and Systems, Requirements for

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes

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MIL-STD-109	Quality Assurance, Terms and Definitions
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment, Electronics, Wired and other Devices
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-470	Maintainability Program Requirements (For Systems and Equipments)
MIL-STD-471	Maintainability Demonstration
MIL-STD-726	Packaging Requirements Codes
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-1275	Electrical Circuit, 28 Volt Transient Characteristics for Military Radios
MIL-STD-2073-2	Packaging Requirement Codes

HANDBOOKS

MIL-HDBK-472	Maintainability Predictions

2.1.2 Other government documents and drawings. The following other Government documents and drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. Marine Corps.

755002B9017	Cable Assembly, Power, Electrical
755002B9018	Cable Assembly, Special Purpose, Electrical
755002B9019	Cable Assembly, Radio Frequency
755002B9100	Amplifier, Power Supply AM-7152/GRC-213

U.S. Army

A3023140	Installation Control Drawing
A3023150	Mounting Base, Electrical MT-6234/GRC-213

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A3023160	Antenna Switching Unit, SA-2365/GRC-213
A3023313	Cable Assembly, Radio Frequency
SM-D-415550	Cable Assembly, Power, Electrical
SM-D-415551	Cable Assembly, Power, Electrical
SM-D-415553	Cable Assembly, Special Purpose, Electrical

DOCUMENTS

SEL FORM 1183	Systems Safety Design Verification Checklist

(Copies of Specifications, standards, handbooks, drawings and other government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Item definition. The Amplifier, Power supply AM-7152/GRC-213 (herein also called amplifier-power supply) and the Adapter Kit, Vehicle MK-2154/GRC-213 (herein also called vehicular adaptor) shall interface with the AN/PRC-104 Radio Set, the AN/VIC-1(V) Intercommunication Set, and Radio Sets AN/VRC-43, 44, 46, 53, and 64. The amplifier-power supply shall also provide loudspeaker output with adjustable squelch. Power lines and interface lines shall have EMI filtering and conditioning to protect interfacing units from transients and RFI. The intended use includes, but is not limited to, M151A2 utility jeep, M561 (Gamma Goat) truck, M113 and M577 Armored Personnel Carrier in all types of weather and terrain which dictates the environmental requirements specified herein. The vehicular adapter shall consist of Switching Unit, Antenna SA-2365/GRC-213 (herein also called antenna switching unit) and Mounting Base, Electrical Equipment MT-6234/GRC-213 (herein also called mounting base). The antenna switching unit shall permit switchable selection of whip or dipole antenna, and selection of necessary antenna loading elements. The mounting base shall reduce shock and vibration levels to the AN/PRC-104 sufficiently such that no degradation of performance of the AN/PRC-104 shall result. The AN/PRC-104, herein referred to as PRC-104, shall consist as a minimum of Receiver- Transmitter RT-1209/URC (herein also called receiver-transmitter) and Radio Frequency Amplifier AM-6874/PRC-104. The combination of the amplifier-power supply and the vehicular adapter (without the AN/PRC-104

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unless otherwise specified) is herein also referred to as equipment. The combination of the vehicular adapter and appropriate cables and bolts constitutes an installation kit.

3.1.1 First article. When specified in the contract or purchase order, and prior to submission of any complete production items for acceptance, the contractor shall furnish first articles for determination of conformance to this specification. Examination and tests shall be those specified herein. Approval of the first articles by the procuring activity shall not relieve the contractor of his obligation to supply vehicular adapters conforming to this specification. Any changes or deviations of the production items from the first articles shall be subject to the approval of the contracting officer.

3.1.2 Interface and cable definition.

3.1.2.1 Amplifier-power supply to receiver-transmitter interface. A removable/replaceable jumper cable (USMC P/N 755002B9018 or equivalent), shall mate physically and electrically with the 5 pin audio connector on the receiver-transmitter and with the 6 pin audio connector on the amplifier-power supply. This cable is supplied with the equipment. This interface is defined in Table III.

3.1.2.2 Amplifier-power supply to AM-6874/PRC-104. A removable/ replaceable jumper cable (USMC P/N 755002B9019) shall mate physically and electrically with the DC power connector P1 of the AM-6874/PRC-104 and the radio power output connector J104 of the Amplifier-Power supply. This cable is supplied with the equipment. This interface is defined in Table IV.

3.1.2.3 Amplifier-power supply to primary input power supply. Interface between the primary input power supply and the 4-pin connector J101 on the amplifier-power supply shall be made with power cable CX-4720 (U.S. Army drawing number SM-D-415550) or equivalent, and is part of the installation kit.

3.1.2.4 Amplifier-power supply to auxiliary output power. Primary output power (unprotected) interface from the 4-pin power connector J103 on the amplifier-power supply to Radio Sets AN/VRC-43, 46, 53 and 64 shall be made with power cable CX-4721 (U.S. Army drawing number SM-D-415551) or equivalent, and is part of the installation kit.

3.1.2.5 Amplifier-power supply to VIC-1(V) intercommunication system. Female receptacle J102 shall provide interface to the AM-1780 of the VIC-1(V) vehicular intercommunication system by means of a CX-4723 cable (U.S. Army drawing number SM-D-415553) or equivalent, and is part of the installation kit. This interface is defined in Table V.

3.1.2.6 Amplifier-power supply to loudspeaker LS-454/U or H-250/U Handset. Audio receptable J1 provides interface to Loudspeaker LS-454/U or Handset H-250/U. This interface is defined in Table VI.

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3.1.2.7 Power amplifier-coupler AM-6874 to antenna switching unit SA-2365. A removable/replaceable 50 ohms RF cable (USMC P/N 755002B9017) with male BNC connectors on both ends shall mate with J1 on AM-6874 amplifier coupler and J201 on the antenna conditioner. This cable is supplied with the equipment.

3.1.2.8 Antenna switching unit SA-2365 to antenna base AB-652B. A special purpose RG-114 cable with male N type connectors on each and (U.S. Army drawing number A3023313) shall mate with J202 on the Antenna Switching Unit and AB-652B antenna base. This cable is part of the installation kit.

3.1.2.9 Antenna switching unit SA-2365 to dipole type antenna. A BNC female connector at the rear of the antenna conditioner J203 shall provide connection to a dipole type antenna. The appropriate cable is supplied with the AN/PRC-104A as part of its antenna kit.

3.2 Performance.

3.2.1 Amplifier-power supply. The amplifier-power supply shall meet the requirements herein when operated from a voltage source of 24.5 to 32 VDC that meets the requirements of MIL-STD-1275. In addition, the amplifier-power supply shall allow the AN/PRC-104 to operate without degradation of its specified performance when the input power source meets the requirements of MIL-STD-1275 for normal operation.

3.2.1.1 Power supply.

3.2.1.1.1 Input power. When operated with supply voltage of 26.5 VDC and with an AN/PRC-104 installed, the input power shall not exceed 200 watts in any mode of operation.

3.2.1.1.2 Reversed polarity. The amplifier-power supply unit and associated receiver-transmitter shall be protected from reversed polarity primary input power. Blowing of a fuse or circuit breaker trip is allowed for this protection. The chassis shall be negative.

3.2.1.1.3 Protected power output. The amplifier-power supply shall provide the AN/PRC-104 with a nominal 100 watts of protected power in compliance with MIL-STD-1275.

3.2.1.1.4 Single fault protection. The amplifier-power supply shall protect the AN/PRC-104 from damage when the power source has a single fault as defined by MIL-STD-1275.

3.2.1.1.5 Transient protection. The amplifier-power supply shall not permit a positive transient at the input to the AN/PRC-104 to exceed 42V when the power source meets the requirements of MIL-STD-1275.

3.2.1.1.6 Maximum protected output current. The amplifier-power supply shall be capable of providing the AN/PRC-104 with up to 5.5 amperes of protected DC current.

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3.2.1.1.7 Overload protection. With short circuit at the protected output power connector no damage to transistor or devices other than an open fuse or circuit breaker trip shall result.

3.2.1.1.8 Auxiliary power output. The amplifier-power supply shall provide unprotected primary power to the auxiliary connector which shall drain a maximum current of 10 amps. This output shall be connected in parallel to the input supply line from the vehicle battery.

3.2.1.2 Loudspeaker amplifier.

3.2.1.2.1 Loudspeaker. The loudspeaker shall be a 2 watt, permanent magnet, type LS-454/U, resistant to fungus, gunblast and immersion, per MIL-L-12632.

3.2.1.2.2 Speaker on/off control. An on/off switch shall be provided to turn the speaker on and off.

3.2.1.2.3 Volume control. The control of the headphone output, the speaker output, and the monitor output to the AN/VIC-1(V) Intercommunication Set, shall be provided by the audio volume control of the AN/PRC-104 radio.

3.2.1.2.4 Audio amplifier input. AF input, from the receiver-transmitter, RT-1209, shall be a minimum of 2 mW into 500 ohms. Distortion of the RT output shall be 5 percent maximum. The loudspeaker volume is controlled by the RT volume control.

3.2.1.2.5 Amplifier output. Minimum AF output over the 300 to 3000 Hz range shall be 1 watt into a 600 ohm loudspeaker. Total harmonic distortion shall not exceed 12 percent.

3.2.1.2.6 Frequency response. The audio response of the loudspeaker amplifier output shall be within $\pm$3 dB of 1000 Hz reference from 300 to 3000 Hz.

3.2.1.3 Monitor amplifier.

3.2.1.3.1 Monitor output. The monitor output, into a 150 ohm resistive load, shall be 0.16 Vrms minimum and 0.31 Vrms maximum.

3.2.1.3.2 Frequency response. The audio frequency response shall be within $\pm$2 dB of 1000 Hz reference from 300 Hz to 3000 Hz.

3.2.1.3.3 Output impedance. The output impedance of the monitor shall be 1800 ohms minimum.

3.2.1.4 Headphone amplifier.

3.2.1.4.1 Headphone output unmuted. The headphone (unmuted) output shall supply a minimum of 80 milliwatts into a 600 ohm resistive load. The total harmonic distortion shall not exceed 10 percent.

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3.2.1.4.2 Frequency response. The audio frequency response shall be within ±2 dB of 1000 Hz reference from 500 Hz to 3000 Hz.

3.2.1.4.3 Output impedance. The output impedance of the headphone amplifier circuit shall be less than 43 ohms.

3.2.1.5 Squelch. Squelched/unsquelched AF output modes shall be applicable to both the loudspeaker and the VIC-1 Intercom System. The squelch system shall provide maximum discrimination against unsquelching on non-voice signals such as noise, RTTY, or telegraphy by recognizing the syllabic low frequency content of voice signals.

3.2.1.5.1 AF reduction. In squelched condition the AF output reduction shall be a minimum of 30 dB at 300 Hz.

3.2.1.5.2 Squelch sensitivity. The AF output shall unsquelch with voice signals having a minimum SINAD of 10 dB. A squelch sensitivity control shall be provided to allow the squelch/unsquelch threshold to be increased to a 20 dB SINAD. A squelch on/off switch shall also be provided.

3.2.1.5.3 Squelch hysteresis. After unsquelching, the unsquelched condition shall be retained for a SINAD reduction of up to 2 dB.

3.2.1.5.4 Attack and release times. The unsquelch attack time shall be a maximum of 35 milliseconds. The release time to the squelched condition shall be delayed at least 300 milliseconds after termination of the voice signal, but not more than 3 seconds.

3.2.1.6 Transmitter audio output. The amplifier-power supply shall provide the AN/PRC-104 transmitter audio input an audio signal of -56 dBm nominal into a 150-ohm load.

3.2.1.7 Retransmit output. When no signal is applied to the amplifier unit the retransmit contact on the squelch relay shall be open. When the signal applied to the amplifier unit includes the syllabic low frequency content of voice signals the squelch shall break and the retransmit contact on the squelch relay shall close and produce a ground on the retransmit output (pin T on J102).

3.2.1.8 Muting. When the muting terminal (J103 pin D) is grounded the audio output of the associated loudspeakers shall be muted. The loudspeaker audio output reduction shall be a minimum of 30 dB.

3.2.1.9 Transmitter push-to-talk control. When the push-to-talk (PTT) control terminal (J102S) is grounded the AN/PRC-104 will transmit (if its mode control switch allows transmit operation) and the associated loudspeaker will mute.

3.2.1.10 Electromagnetic compatibility (EMC). Filtering of the interface leads (power, audio, and control) listed in Tables III, IV, V and at the audio connectors shall be sufficient to avoid radio frequency interference in installations and communication systems listed in Table VII.

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3.2.2 Radio set AN/PRC-104. (Properly connected to amplifier-power supply, mounted on vehicular adapter).

3.2.2.1 Input voltage. Input voltage to the AN/PRC-104 shall be 20.0 VDC minimum to 32 VDC maximum.

3.2.2.2 Output power. RF power output of the AN/PRC-104 into a 50 ohm resistive dummy load shall be 20 W CW or PEP +2, -3 dB over the frequency of 2.000 to 29.999 MHz.

3.2.2.3 Hum and ripple. Hum and ripple shall be at least 40 dB below the output level of a single tone modulating the transmitter to rated peak envelope power.

3.2.2.4 Receiver sensitivity. The output signal-plus-noise-plus-distortion to noise-plus-distortion ratio (SINAD) shall be at least 10 dB when a radio frequency (RF) signal of 1.4 uv open circuit from a 50 ohm source is applied to the receiver input terminals for a 2.000 to 29.999 MHz range.

3.2.3 Antenna switching unit.

3.2.3.1 Impedance matching. The antenna switching unit shall facilitate impedance matching of a 15 foot whip antenna with a 4 foot lead to enable the automatic tuner of the AN/PRC-104 to tune the antennas.

3.2.3.2 Dipole antenna. The antenna switching unit shall provide connection to a dipole type antenna.

3.2.3.3 Antenna selection. The antenna switching unit shall have a rotary switch on the front panel which will enable the selection of the whip or dipole type antenna and the proper matching network.

3.3 Design and construction. The Amplifier, Power Supply AM-7152/GRC-213 and Adapter Kit, vehicle, MK-2154/GRC-213 shall contain all of the necessary circuitry including system control circuits to allow the operation of the AN/PRC-104 as a 20 W HF vehicular radio system.

3.3.1 Production drawings. The Amplifier, Power Supply AM-7152/GRC-213 and Adapter Kit, vehicle, MK-2154/GRC-213 shall conform to drawings as specified below.

3.3.2 Standards of manufacture.

3.3.2.1 Outline and mounting. Installation control drawing no. A3023140 controls the outline and mounting details. Essentially the system shall consist of the following:

a. A mounting base (drawing no. A3023150) with resilient shock absorbers.

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b. An amplifier-power supply (drawing no. 755002B9100) containing associated electronics. The amplifier-power supply shall include clamps capable of securely clamping an AN/PRC-104 to its top surface. All controls shall be front panel mounted. Power connectors and VIC-1 interface connector shall be rear panel mounted.

c. An antenna switching unit (drawing no. A3023160) containing the antenna conditioning network.

3.3.2.2 Weight. The weight of the amplifier-power supply and vehicular adapter, including the AN/PRC-104 shall not exceed 40 pounds.

3.3.2.3 Controls. The following adjustments and controls shall be provided:

Loudspeaker	Off/On
Squelch	Off/On and threshold adjustment (combined on one knob)
Power	Off/On Magnetic circuit breaker
Lamp	Power on control indicator (dimmable)

3.3.2.4 Construction. The amplifier-power supply shall be constructed as a three piece enclosure, consisting of a cover for access to the electronics, a removable front panel, and the main enclosure. Cover and panel screws shall be captive. The antenna switching unit shall be a two piece enclosure, utilizing a housing with an access cover. The cover screws shall be captive.

3.3.2.5 Reliability. Reliability of operation shall be considered of prime importance in the design and manufacture of the amplifier-power supply and vehicular adapter. The contractor shall employ all methods possible in the process of design, development, and manufacture which will assure quality and maximum reliability consistent with the state-of-the-art. In the functional application of parts, adequate factors of safety shall be provided by suitable derating from the part specification values in order to assure high equipment reliability under all applicable service conditions. The design shall include all possible features which will result in reliable and stable operation and hence reduce the requirements for maintenance by highly skilled personnel. The equipment in a radio set configuration, including AN/PRC-104, AM-7152 and MK-2154 Adapter Kit shall have a combined MTBF of 1600 hours, when subjected to test conditions of Test Level E (modified to a temperature range of -32 to +43°C) and the requirement of Test Plan XXI of MIL-STD-781 (Transmitter duty cycle to be 1:9).

3.3.2.6 Maintainability. Design criteria for the maintainability of the amplifier-power supply and vehicular adapter and support equipment shall be in accordance with MIL-STD-470, MIL-STD-471 and MIL-HDBK-472.

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3.3.2.7 Performance. The equipment shall be capable of the following performance, with the AN/PRC-104 properly attached.

3.3.2.7.1 Operation. The equipment shall operate in wheeled or tracked vehicles, either stationary or in motion, powered by the normal primary DC power source of the vehicle.

3.3.2.7.2 Compatibility. The equipment shall be operationally compatible with the AN/PRC-104, vehicles and communication systems as listed in Table VII. Audio input/output shall be compatible with the H-250/U handset and the KY-65 secure voice equipment.

3.3.2.7.3 Standard conditions. Unless otherwise specified, the following standard test conditions shall apply.

3.3.2.7.3.1 Atmosphere. Prevailing factory conditions of temperature, humidity, pressure.

3.3.2.7.3.2 Input voltage. 26.5 +3.5, -2.0 VDC.

3.3.2.7.4 Service conditions. The equipment shall be capable of operation under the following service conditions: The equipment is meant to be the amplifier-power supply and the vehicular adapter without the AN/PRC-104, unless the AN/PRC-104 is specifically mentioned. (AN/PRC-104 performance limits are defined in MIL-R-29418).

3.3.2.7.4.1 Low temperature.

3.3.2.7.4.1.1 Operating. The equipment shall withstand exposure to ambient air temperatures as low as -46°C (-50°F) while operating, when tested in accordance with 4.2.2.1. Verification shall be by measurement.

3.3.2.7.4.1.2 Storage and transportation. The equipment shall withstand exposure to ambient air temperatures as low as -62°C (-80°F) when tested in accordance with 4.2.2.1. Verification shall be by measurement.

3.3.2.7.4.2 High temperature.

3.3.2.7.4.2.1 Operating. The equipment shall withstand exposure to ambient air temperatures as high as +71°C (+160°F) while operating, when tested in accordance with 4.2.2.1. Verification shall be by measurement.

3.3.2.7.4.3 Humidity. The equipment shall operate without degradation in specified performance during and after exposure to extreme high humidity levels (0-100%) as encountered in tropical areas, without physical damage, including conditions where condensation occurs on the exterior of the equipment. There shall be no deterioration or corrosion of finishes or metals when tested in accordance with 4.2.2.2. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface. Verification shall be by inspection.

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3.3.2.7.4.4 Altitude. The equipment shall operate without degradation in specified performance at altitudes over the range of sea level to 10,000 feet and shall not be damaged by exposure to altitudes of 50,000 feet. The rate of change of altitude shall not exceed 2000 feet per minute. The amplifier-power supply shall meet the altitude requirement when tested in accordance with 4.2.2.4. Verification shall be by measurement.

3.3.2.7.4.5 Vibration and shock. The equipment shall withstand vibration and shock induced during vehicular transport upon military vehicles over all types of roads and cross country terrain. In addition the equipment shall be resistant to high level shock caused by ballistic impact, air explosions, and similar combat field conditions, when tested in accordance with 4.2.2.5. Verification shall be by measurement.

3.3.2.7.4.5.1 Amplifier-power supply, unloaded. The equipment (without the AN/PRC-104 mounted on it) shall not exhibit damage and shall operate without degradation in specified performance after exposure to vibration when tested in accordance with 4.2.2.5. Verification shall be by inspection.

3.3.2.7.4.5.2 Amplifier-power supply, loaded. When the AN/PRC-104 is properly mounted on the amplifier-power supply the requirements of 3.3.2.7.4.5.1 shall be met during and after the test of 4.2.2.5. Verification shall by by inspection.

3.3.2.7.4.5.3 AN/PRC-104. During the test of requirement 3.3.2.7.4.5.2 the requirement for power output (3.2.2.2) shall be met, the requirement for hum and ripple (3.2.2.3) shall be met and the requirement for receiver sensitivity (3.2.2.4) shall be met. Verification shall be by measurement.

3.3.2.7.4.6 Shock, bench handling. The equipment and its subassemblies shall not exhibit damage or subsequently fail to operate without degradation in specified performance when tested as specified in 4.2.2.6.

3.3.2.7.4.7 Shock, drop. The equipment as packaged for shipment, shall not exhibit damage or subsequently fail to operate without degradation in specified performance when tested as specified in 4.2.2.7.

3.3.2.7.4.8 Bounce and shock. The equipment shall withstand the requirements of Table I without physical damage when tested in accordance with 4.2.2.8. Verification shall be by measurement and inspection.

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TABLE I. Bounce and Shock.

Requirements	Test Para	Performance After Test
(a) Bounce	4.2.2.8.1	Specified performance (Note A). No physical damage, except for minor paint abrasion.
(b) Shock; ballistic	4.2.2.8.2	Specified performance (Note A). No physical damage, except for minor paint abrasion.

Note A: After completion of the tests, the equipment (including the AN/PRC-104) shall meet the specified requirements of 3.2.

3.3.2.7.4.9 Sand and dust. The equipment shall operate within the specified tolerance limits for measured parameters during and after prolonged exposure to sand particles with wind speeds of 1750 fpm +250 fpm, at low humidity as simulated by test in accordance with 4.2.2.9. The equipment shall show no evidence of damage or abrasion or clogging from wind blown sand particles which affects the operation of the equipment.

3.3.2.7.4.10 Leakage (immersion, covers and seals). The amplifier-power supply and antenna switching unit shall not exhibit evidence of leakage after immersion in water for 2 hours at a depth of 0.9144 meters +10.77 cm - 0 cm (3 feet +5 inches - 0 inches) in accordance with 4.2.2.10.

3.3.2.7.4.11 Fungus. The equipment (including accessories) shall provide no nutrients in material, coating or contaminent form or support fungal growth when tested in accordance with paragraph 4.2.2.11. The covers shall be open and the equipment shall not be required to operate after fungus exposure. Only inherently fungus resistant grades of materials per requirement 4, MIL-STD-454 shall be used. After 28 days, based on visual examination, the equipment or samples shall show no more then sparse microbial growth with restricted tuberal growth development in an area 10% or less of the total area and no more than six unrelated minute colonies, with mycelial development in non-critical areas. Critical areas include: terminal spacing, printed circuit boards, etc. The equipment shall fail if it shows more than the growth specified above.

3.3.2.7.4.12 Salt fog. The equipment shall be resistant to the corrosive effects of a salt fog atmosphere when tested in accordance with 4.2.2.12. There shall be no corrosion of finishes or metals. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surfaces. There shall also be no clogging or binding of moving mechanical parts. Verification shall be by visual and mechanical inspection.

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3.3.2.7.5 Storage and transportation conditions. The equipment shall be capable of being stored and transported under the following conditions without degradation.

a. Ambient Temperature Range: -62°C to +71°C.
b. Altitude: Sea level to 50,000 feet.

3.3.2.7.6 Control life. All controls shall be functional end-to-end without failure for no less than 10,000 cycles.

3.3.2.7.6.1 Control and stops. All controls identified as having end stops shall be capable of withstanding repeated force in each direction of 48 inch-ounces without permanent deformation to end stops.

3.3.2.8 Marking. Marking shall conform to Requirement 67 of MIL-STD-454. Panel marking for switch positions shall be accurately aligned with switch detents.

3.3.2.9 Finish, protective. The equipment shall be finished in accordance with MIL-C-46168(ME) Type 2, Forest Green and the equipment drawings. This includes finish of hardware, such as handles, hinges, screws, with necessary touch-up after mounting and testing. Pretreatment film on all exterior surfaces shall be per MIL-F-14072 with primer per MIL-P-23377.

3.3.2.10 Parts, materials and processes. Parts, materials and processes approved in MIL-E-16400 shall be given first consideration. Parts shall be selected from MIL-P-11268 wherever possible. All nonstandard parts shall be approved by the procuring activity before delivery of units. Particular attention shall be paid to the design, application and selection of parts, including solid state devices, as specified in MIL-E-16400. Parts and solid state units shall be used in such a manner that over-stressing, or operation near maximum rating, shall not be allowed to occur.

3.3.2.11 Duty cycle. The amplifier-power supply and the vehicular adapter shall be capable of continuous use.

3.3.2.12 Serial numbers. The following items shall have serial numbers:

Amplifiers, power supply AM-7152/GRC-213
Switching Unit, Antenna SA-2365/GRC-213
Mounting Base, Electrical Equipment MT-6234/GRC-213

3.3.2.13 Burn-in. The amplifier power supply shall be able to withstand the burn-in test as specified in Section 4 prior to acceptance for delivery.

3.3.3 Workmanship. Workmanship shall be in accordance with Requirement 9 of MIL-STD-454.

3.3.4 Covers and seals. Cover seals or sealing surfaces shall be capable of 25 open and close cycles without loss of leakage resistance.

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3.3.5 Human performance/human engineering. Human engineering shall conform to Requirement 62 of MIL-STD-454.

3.3.6 Interchangeability. Provisions for interchangeability are required and shall be in accordance with Requirement 7 of MIL-STD-454. Like units, assemblies, subassemblies, modules and replaceable parts shall be physically, electrically and functionally interchangeable without modification of such items or compatible equipment.

3.3.7 Safety. Personnel safety hazards shall be kept to a minimum in accordance with Requirement 1 and 8 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. When specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein subject to government approval. When specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractors overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.1.1 Quality program requirement. The contractor shall provide and maintain a written quality program acceptable to the Government for supplies and services covered herein (see 6.2.4). The quality program shall be in accordance with MIL-Q-9858.

4.1.1.2 Quality plan. A quality plan, acceptable to the equipments covered herein shall be prepared if specified in a CDRL (DD Form 1423) for Government approval (see 6.2.4).

4.1.1.3 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will include (a) surveillance of the operations to determine that practices, methods, and procedures of the written inspection system are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to

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promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.1.4 Test procedures. When specified in the contract or purchase order, the contractor shall prepare test procedures for Government approval covering first article and quality conformance inspection (see 6.2.4).

4.1.1.5 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.1.2 Special tests and examinations.

4.1.2.1 Classification of inspections and tests. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.1.2.1.3).
b. Production Inspection (see 4.1.2.1.4).
c. Sample Inspection (see 4.1.2.1.5).
d. Environmental Inspection (see 4.1.2.1.6).

4.1.2.1.1 Test plans. When specified in the contract or purchase order, detailed test plans describing in detail the first article production and sample testing shall be prepared by the contractor in accordance with the requirements of MIL-STD-831 and submitted to the contracting officer for approval prior to initiation of testing (see 6.2.4). Approval of the test plans is required prior to the commencement of testing. The test plans shall include all test procedures deemed necessary and practicable by the contractor to demonstrate compliance with this specification.

4.1.2.1.2 Tests. Table VIII sets forth testing requirements for first article, production, sample and environmental conditions. These tests are further described in 4.1.2.1.6 and 4.2.3.

4.1.2.1.3 First article inspection. The contractor shall furnish units for first article inspection. The Government representative present will select which items are to be subjected to each test. The requirement of paragraph 4.1.2.1.4.1 applies as a prerequisite to performing first article inspection tests.

4.1.2.1.3.1 First article inspection report. A first article inspection report shall be submitted to the Government covering the results of the contractors first article inspection (see 6.2.4), in accordance with the CDRL.

4.1.2.1.3.2 First article tests. First article inspection shall be as indicated in Table VIII.

4.1.2.1.4 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the

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omissions and errors of the production process, such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment and tests which detect hidden defects of material. Production inspection shall consist of the tests and examinations approved in accordance with 4.1.1.4 and shall include the requirements of Table VIII as specifically indicated. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order shall be in accordance with MIL-STD-105. Each lot shall be subjected to sampling inspection, utilizing the procedures and the general inspection levels of MIL-STD-105 and AQL of 1 percent for all electrical inspections. The AQL for visual and mechanical inspections shall be 10 defects per hundred units for major defects and 15 defects per hundred units for minor defects (major and minor defects are defined in MIL-STD-105 and MIL-STD-252. Defects in all inspections other than visual and mechanical shall be considered major. Inspections other than visual and mechanical may be performed in any order.

4.1.2.1.4.1 Group A inspections. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in paragraph 4.1.2.1.5. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures and the general inspection levels of MIL-STD-105 and an AQL of 1 percent for all electrical inspections. The AQL for visual and mechanical inspections shall be 10 defects per hundred units for major defects and 15 defects per hundred units for minor defects (major and minor defects are defined in MIL-STD-105 and MIL-STD-252). Defects in all inspections other than visual and mechanical shall be considered major. Group A inspection specified in paragraph 4.1.2.1.5 shall be performed in any order.

4.1.2.1.4.2 Group B inspections. Group B inspections shall be performed on inspection lots that have passed Group A inspection. Group B inspections are specified in paragraphs 4.1.2.1.4, 4.1.2.1.5 and Table II. This inspection shall conform to the special inspection levels of Table I of MIL-STD-105. The Group B AQL shall be 4 percent and the inspection level shall be S-4 (per MIL-STD-105)

4.1.2.1.4.3 Inspection of parts, subassemblies, and items. Visual and mechanical inspections shall be performed to determine conformance to the specified requirements of MIL-E-16400 prior to inspection of preparation for delivery.

4.1.2.1.4.3.1 Mechanical inspection. The equipment shall be given a thorough mechanical and visual inspection to determine that the quality of all materials and workmanship is in compliance with the requirements specified herein and in MIL-E-16400. Particular attention shall be given to the following:

a. Completeness

b. Name plates, identification markings and labels.

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c. Ease of operation of connector mating, thumb screws, controls and switches and any other part intended for adjustment, movement or operation.

d. Finishes.

e. Welded joints.

f. Check of solder joints.

g. The fit of components in their respective positions.

h. Check of mounting means.

i. Check of safety features.

j. Fastening and securing devices of parts.

k. Accessibility of components and parts for servicing.

l. Cable run between components including plugs and receptacles.

m. Fit of sealing gasket on cover and wear on gasket.

n. Check of all connector fittings.

o. Test points.

p. Overall dimensions.

q. Weight.

r. Other visual defects.

4.1.2.1.5 Sample inspection. Sample inspections shall be performed on inspection lots that have passed production inspection. Sample inspection shall conform to Table VIII and shall conform to the special inspection levels of Table I of MIL-STD-105. The sample inspection AQL shall be 4 percent and the inspection level shall be S-4 (per MIL-STD-105).

4.1.2.1.6 Environmental inspection. Sample rates for environmental inspection shall apply to the production rate of the basic radio set and shall not exceed the cumulative total of lot samples defined by the following sampling rate.

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Quantity Produced (per lot)	Quantity to be Tested
500	1 for Type I 1 for Type II 1 for Type III
Production less than 500	1 for Type I 1 for Type II 1 for Type III

(Dust and Fungus tests are deleted after first successful test of production samples)

4.1.2.1.6.1 Nonconforming environmental sample units. In the event a sample unit fails any type of environmental inspection, actions required shall be as specified in the contract or purchase order.

4.1.2.1.6.2 Reinspection of conforming environmental sample units. Unless otherwise specified in the contract (see 6.2.1), sample units which have been subjected to, and passed, environmental tests may be accepted on the contract provided they are resubjected to, and pass, production inspection after repair of all damage.

TABLE II. Environmental inspection (group C).

Inspection	Paragraph
Type I	
Temperature (High and Low)	4.2.2.1
Leakage (Immersion)	4.2.2.10
Bench Handling	4.2.2.6
Type II	
Altitude	4.2.2.4
Vibration	4.2.2.5
Vehicular Bounce	4.2.2.8.1
Shock, Ballistic	4.2.2.8.2
Shock, Drop	4.2.2.7
Humidity	4.2.2.2
Type III	
Dust	4.2.2.9
Salt Fog	4.2.2.12
Fungus	4.2.2.11

4.2 Quality conformance inspections.

4.2.1 Safety engineering. A visual inspection shall be performed to verify compliance with the requirements of 3.3.7. SEL FORM 1183 must be completed as part of First Article Testing.

4.2.2 Environmental tests.

4.2.2.1 Temperature. The equipment shall be tested in accordance with MIL-STD-810.

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4.2.2.1.1 Low temperature. The equipment shall be subjected to the test of MIL-STD-810, Method 502.2, Procedure I. Lower the internal chamber temperature in Step 2 to -62°C (80°F) for a period of not less than two hours following stabilization of the equipment, or until 48 hours have elapsed. In Step 5 of Procedure I, perform Procedure II-Operation (paragraph II-3.2 of Method 502.2). The chamber temperature in Step 1 of Procedure II shall be adjusted to -45.6°C (-50°F) operating input power of +24 VDC to the unit. The temperature of the chamber shall then be raised to -40°C (-40°F) and allowed to stabilize. Operational tests shall be performed with +20 VDC and +32 VDC input power to the unit. The requirements of 3.3.2.7.4.1 shall be met.

4.2.2.1.2 High temperature. The equipment shall be subjected to the test of MIL-STD-810, Method 501.2, Procedure II for cycling exposure. Testing shall the continuous during transition form low temperature, operating, to high temperature, operating. Starting at a temperature of -45.6°C (-50°F) and an input power to the unit of 24 VDC, the temperature shall be first increased to -40°C (-40°F) and then by 20°F degree steps in Step 12 to +49°C (+120°F), with operational tests and input power +20 VDC and +32 VDC, at each gradient. Maintain internal chamber temperature for six hours at +49°C (+120°F). Raise the internal chamber temperature to +71°C (+160°F) within a time period of one hour and maintain that temperature for four additional hours. Perform an operational test with +20 VDC and +32 VDC. Lower the internal temperature to +49°C (+120°F) within a period of one hour. Repeat the above two steps two additional times for a total of three 12 hour cycles. Step 14 shall be performed as follows: adjust the internal chamber temperature until temperature stabilization of the test item is reached. Operate the test item until the item is stabilized. The requirements of 3.3.2.7.4.2 shall be met.

4.2.2.2 Humidity. The equipment shall be subjected to the test of MIL-STD-810, Method 507.2, Procedure II. Operational tests shall be performed at the 5th and 10th cycles. The requirements of 3.3.2.7.4.3 shall be met.

4.2.2.3 Rain. The equipment shall be subjected to the test of MIL-STD-810, Method 506.2, Procedure I. In Step 1, the test item shall be exposed to a simulated rain at a rate of 2 $\pm$0.5 inches per hour for 10 minutes. The rate of rainfall shall then be raised to 5 $\pm$1 inches per hours and held at this rate for 5 minutes. The rate shall then be reduced to 2 $\pm$0.5 inches per hour for the next 15 minutes. Step 2 shall be changed as follows: starting 5 minutes after the initiation of the rain, the wind source shall be turned on and adjusted to produce a horizontal wind velocity of 40 miles per hours (3,500 feet per minute). The wind source shall be turned off. The test item shall be operated in Step 3. The requirements of 3.3.2.7.4.3 shall be met.

4.2.2.4 Altitude. The equipment shall be subjected to the test of MIL-STD-810, Method 500.2, Procedure II (operating). The chamber pressure in Step 2 shall be adjusted to 508mm of Hg (20 inches of Hg or 10,000 feet above sea level) at a rate not to exceed 2,000 fpm. The equipment shall also be subjected to the test of MIL-STD-810, Method 500.2, Procedure I (non-operating). For Step 2, adjust the chamber pressure to 95mm of Hg (3.75 inches of Hg or 50,000 feet above sea level), at a rate not to exceed 2,000 fpm. The requirements of 3.3.2.7.4.4 shall be met.

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4.2.2.5 Vibration. The equipment shall be mounted by normal means and tested while operating. The test item shall be vibrated along each axis in accordance with the applicable test levels and frequency range, from the applicable figures. The frequency of applied vibration shall be swept over the specified range logarithmically in accordance with Figure 1. The specified sweep time is that of an ascending plus a descending sweep and is twice the ascending sweep shown on Figure 1 for the specified range. Test levels shall be according to Figure 2. The test time schedule shall be sinusoidal cycling time of 30 minutes/1000 miles for a maximum cycle time of 5.5 hours and a sweep time of 12 minutes for 5 Hz-200-5 Hz. When test item resonances below 5 Hz are measured or expected the test curves shall be extended to 2 Hz and the sweep time shall be 18 minutes (2-500-2 Hz): 15 minutes (2-200-2-Hz). Test duration shall be 3 hours per axis with a 30 minute dwell at each of the most severe resonances (4 maximum). The requirements of 3.3.2.7.4.5 shall be met. First Article Tests (FAT) shall be conducted with the test item mounted in a representative installation kit specified in the test procedure. The FAT shall not be repeated for each type of installation kit.

a. Resonant frequencies shall be interpreted to include:

   - isolation if any
   - circuit card assembly
   - radio set response

b. Production sample tests may be conducted with the radio set mounted in installation kits selected in proportion to the number of kits produced within the sample interval.

c. Resonance shall be defined as a resonant rise in amplitude of 2:1 or greater relative to the amplitude (g) off resonance. The amplitude may or may not be equal to the input amplitude (g).

d. The severity of resonance will be considered to scale as

$$D - P = G^4 \, ft$$

   where
   G = acceleration, g's
   f = frequency, Hz
   t = time of acceleration

The radio output shall be monitored for vibration sensitive spurious outputs as one indication of resonance.

4.2.2.6 Shock, bench handling. Major subassemblies of the equipment shall be subjected to the requirements of MIL-STD-810, Method 516.3, Procedure VI. The requirements of 3.3.2.7.4.6 shall be met.

4.2.2.7 Shock, drop. The equipment shall be packaged for shipment and the shipping package shall be drop tested from a height of 48 inches, each face, each corner and each edge for a total of 26 times. Perform an operational test on completion of drop testing to determine compliance with 3.3.2.7.4.8 Note A. The requirements of 3.3.2.7.4.7 shall be met.

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4.2.2.8 Shock, bounce and ballistic. The equipment shall be mounted by normal means and shall be loaded with the AN/PRC-104. The assembly will be subjected to non-operating shock test as specified in 4.2.2.8.1 and 4.2.2.8.2. The requirements of 3.3.2.7.4.8 shall be met.

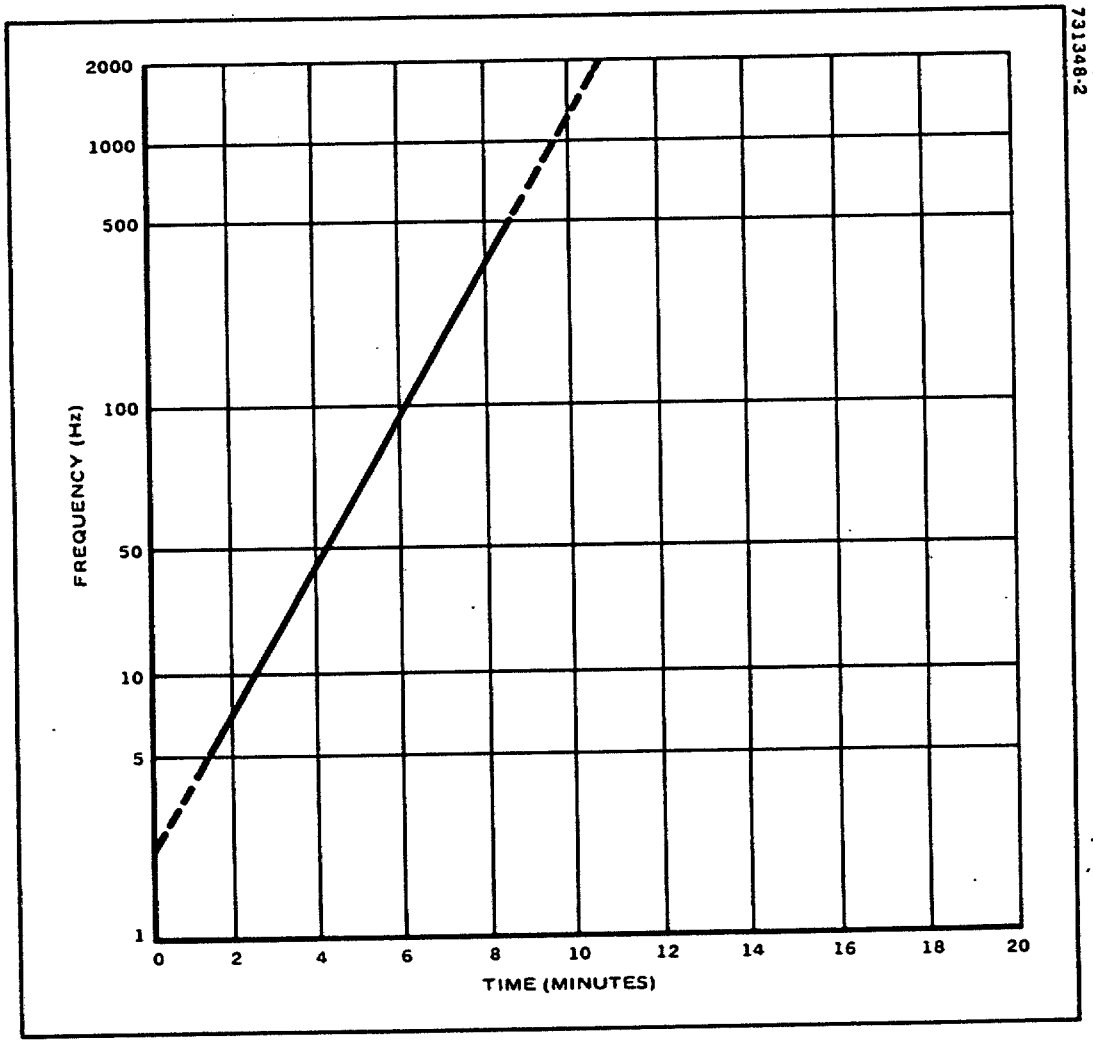
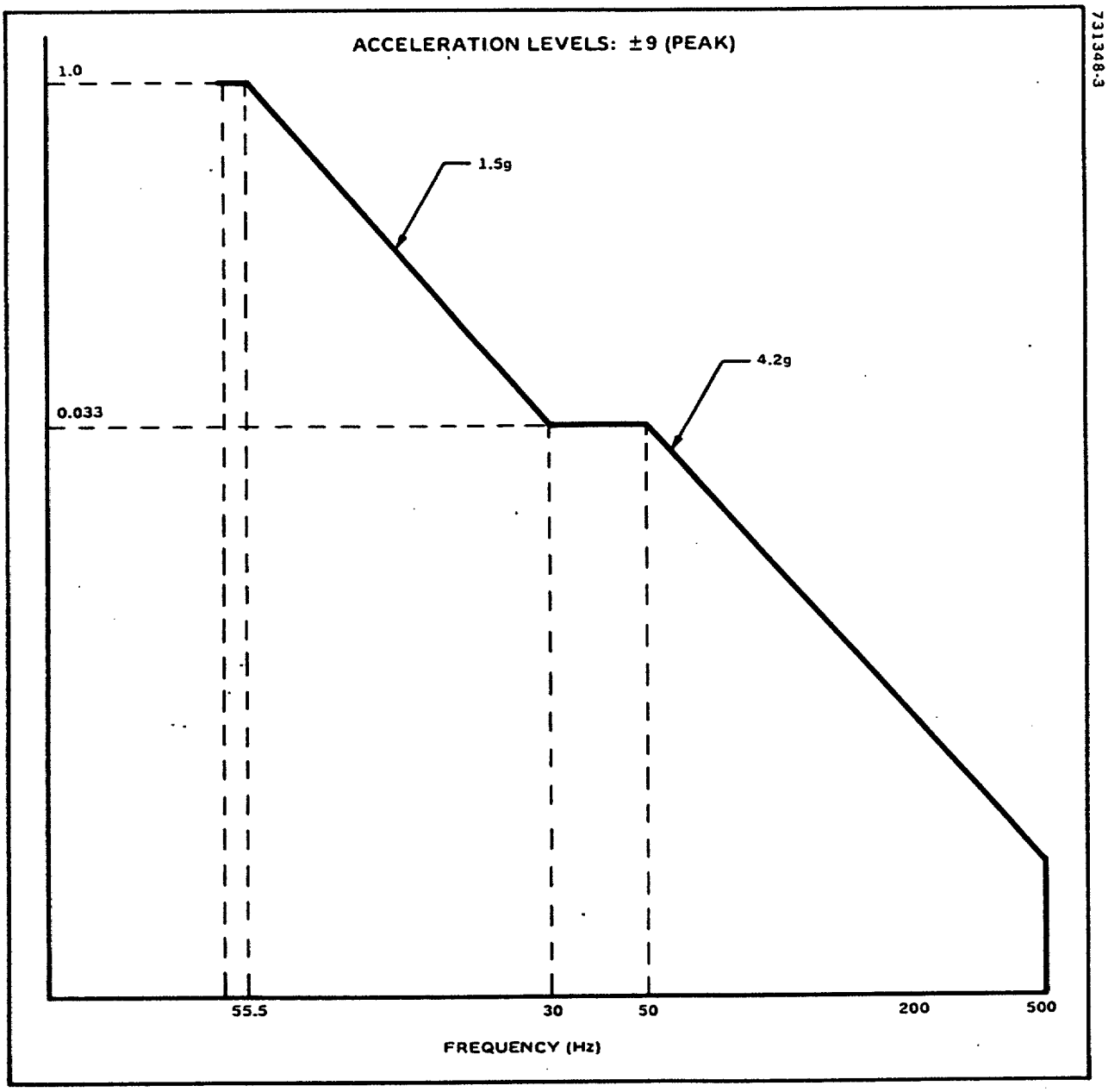


Figure 1. Logarithmic sweep.

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Note: All curves shall be extended to 2 Hz when test item resonances below 5 Hz are expected.

Figure 2. Vibration test curves for equipment Installation in ground vehicles.

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4.2.2.8.1 Loose cargo transport. The equipment shall be tested in accordance with MIL-STD-810, Method 514.3, Procedure II, Test condition I-3.2.3. The test shall be conducted utilizing a package tester capable of 1 inch (double amplitude) displacement and of suitable capacity for testing military equipment. The test bed of the package tester shall be covered with a panel of 1/2 inch plywood secured by nails. The tester shall be adjusted to a speed such that the average value of the random acceleration peaks shall be 7.5 ±2.5 g's. Measurement shall be conducted at the output of a bandpass filter with a band pass of 0.2 to 100 Hz and an attenuation slope of 12 to 18 dB per octave at the 3 dB down point. Infrequent pulses greater than 10 g's need not be used in calculating the average. The test shall be performed for a total of 3 hours. At the end of each 3/4-hour period, the equipment shall be rotated 90 degrees, each time in the same direction. After completion of the test the equipment shall be operationally tested to determine compliance with 3.3.2.7.4.8 Note A.

4.2.2.8.2 Shock test, ballistic. The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" shown in specification MIL-S-901. The equipment, mounted on Mount MT-6234/GRC-213, shall be secured in its normal mounting position to the Platform No. 1 fixture 4 C per figure 6 of MIL-S-901 by means of the same fasteners used for vehicular installation of the equipment. The test shall consist of a total of 9 blows: one each 1-foot blow, 3-foot blow, and 5-foot blow on the back, side, and top of the test plate. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible. After completion of the shock test, the equipment shall be tested to determine compliance with 3.3.2.7.4.8 Note A.

4.2.2.9 Sand and dust. The equipment shall be subjected to the test of MIL-STD-810, Method 510.2, Procedure II. The air velocity in Step 2 shall be 1750 fpm ±250 fpm. The equipment shall be reoriented periodically, such that each side of the equipment is exposed to the sand stream for 90 minutes. Step 5 and 7 shall be omitted. The requirements of 3.3.2.7.4.9 shall be met.

4.2.2.10 Leakage (immersion). The equipment shall be immersed in a tank of water to a depth of 0.9144 meters +10.77cm -0 cm (3 feet +5 inches -0 inches). The water shall be heated to a temperature of +18°C ±5°C (64.4°F ±10°F). The equipment shall be immersed until all evidence of bubbles have disappeared. The equipment shall then be removed and the covers and seals removed and replaced 25 times. The equipment shall then be placed in a chamber and heated to +45°C ±3°C (+113°F ±5°F) and then immersed in the tank. Evidence of leakage shall be observed. The requirements of 3.3.2.7.4.10 shall be met.

4.2.2.11 Fungus. The equipment shall be fungus tested in accordance with MIL-STD-810, Method 508.3, with all gasket sealed covers open. Operation of the equipment after test shall not be required. The requirements of 3.3.2.7.4.11 shall be met.

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4.2.2.12 Salt fog. The equipment shall be subjected to the test of MIL-STD-810, Method 509.2; Procedure I. In Step 2 the procedure shall be constant wetting. Delete Step 3 and 4 (operation). The requirements of 3.3.2.7.4.12 shall be met.

4.2.3 Tests. (All tests shall be performed in accordance with contractor prepared Government approved test procedures; which shall include accept/reject criteria). Tests shall be in accordance with Table VIII.

4.2.3.1 Input power. The equipment shall be operated and tested with an input voltage, of 26.5 ±0.5 VDC to demonstrate compliance with the requirements of 3.2.1.1.1.

4.2.3.2 Reversed polarity. The equipments shall be tested to demonstrate no damage other than a blown fuse or circuit breaker trip for reversed polarity input power. Production sample testing may be implemented so as to eliminate the destruction of fuses.

4.2.3.3 Input voltage. The equipment shall be operated and tested over the input voltage range of 24.5 to 32 VDC to demonstrate compliance with the requirements of 3.2.1.

4.2.3.4 Loudspeaker amplifier. Amplifier tests for frequency response, output power, and distortion shall be made with a 600 ohm resistive dummy load substituting for the loudspeaker.

4.2.3.5 Squelch. As detailed in an approved test procedure, squelch characteristics shall be tested through the interconnected receiver-transmitter with a modulated RF signal source. A voice signal shall be simulated by square wave modulation of the 1 kHz received audio at a 5 Hz rate.

4.2.3.6 Electromagnetic compatibility. No EMC tests are necessary for the vehicular adapter as an isolated unit. Electromagnetic compatibility shall be proven in system tests according to an approved system test plan.

4.2.3.7 Maintainability. The maintainability demonstration requirements of 3.3.2.6 shall use contractor furnished personnel. Maintainability shall be demonstrated as specified in contract or purchase order.

4.2.3.8 Compatibility. As detailed in an approved test plan, interface compatibility per Tables III, IV, V and VI shall be proven by interoperational tests of the interconnected amplifier-power supply, AN/PRC-104, and H-250/U handset. Interface compatibility with he VIC-1 Intercommunication system per Table III, IV, V, and VI shall be verified by voltage measurements at an interface simulation test box according to prescribed control manipulations.

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4.2.3.9 Control life. Control life requirements of 3.3.2.7.6 shall be tested on an assembled amplifier-power supply.

4.2.3.10 Control end stops. The control and stop requirements of 3.3.2.7.6.1 shall be tested on an assembled amplifier-power supply.

4.2.3.11 Burn-in. The burn-in test shall consist of 20 hours of failure-free operation during which all front panel control switches shall be operated and proper operation shall be monitored for each switch position.

4.2.3.12 Interchangeability. Electrical and mechanical interchangeability tests shall be performed at the module and unit level with an interconnected AN/PRC-104 and VIC-1 to establish conformance to the requirements of 3.3.6.

4.2.3.13 Reliability. When specified in the contract or purchase order, the contractor prepared Government approved test shall be performed to establish compliance with the requirements of 3.3.2.5.

4.2.3.14 Workmanship screen. Prior to conducting temperature cycling test, vibration shall be performed on each equipment. The vibration may be performed at the module, drawer, or end item level. All the hardware including cables and connectors shall be exposed to vibration.

4.2.3.14.1 Vibration testing. The vibration shall be random, or subject to procuring activity approval, pseudorandom or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All test items shall be subjected to the vibration conditions of Figure 3. The control accelerometer shall be located next to one of the mounting points of the item under test. Equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 Hz shall be used for the control and analysis of the acceleration spectral density.

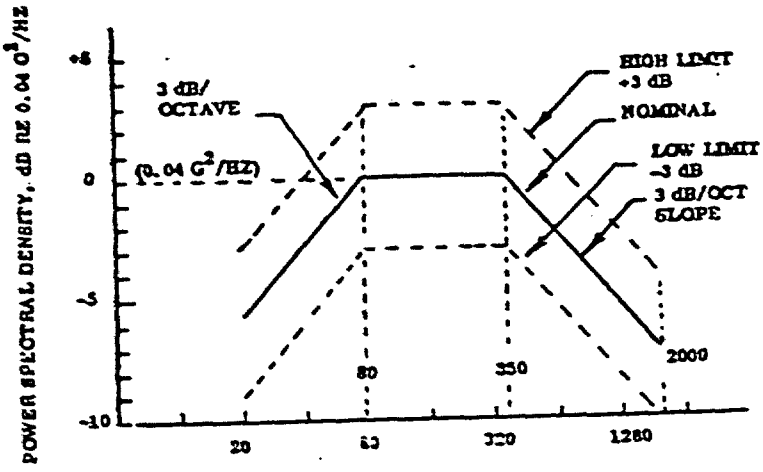


Figure 3. Random vibration curve.

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4.2.3.14.2 Test item. The test item shall be energized during vibration and appropriate input signals applied to serve any abnormal conditions of the output functional characteristics. All failures occurring during test shall be corrected and test resumed.

4.2.3.15 Temperature cycling. Each equipment shall be subjected to the temperature curve in Figure 4. The number of cycles required shall be determined by the electric/electronic parts count and the parts count chart below. Equipment power shall be turned on and off at the indicated times. When practical, the equipment enclosures shall be opened or removed for maximum exposure to the changing temperature. The last temperature cycle shall be failure free. When performance measurements are called for, a functional operating test shall be performed. The dwell time shall be 80 percent of the time required for the largest electrical/electronic part to become temperature stabilized. Temperature stabilization shall be as defined in MIL-STD-810 for the non-operating condition.

Electric/Electronic Parts Count	No. of Temperature Cycles
500 or less	3
501 to 1000	4
1001 to 2000	6
2001 to 3000	8
3001 and up	10

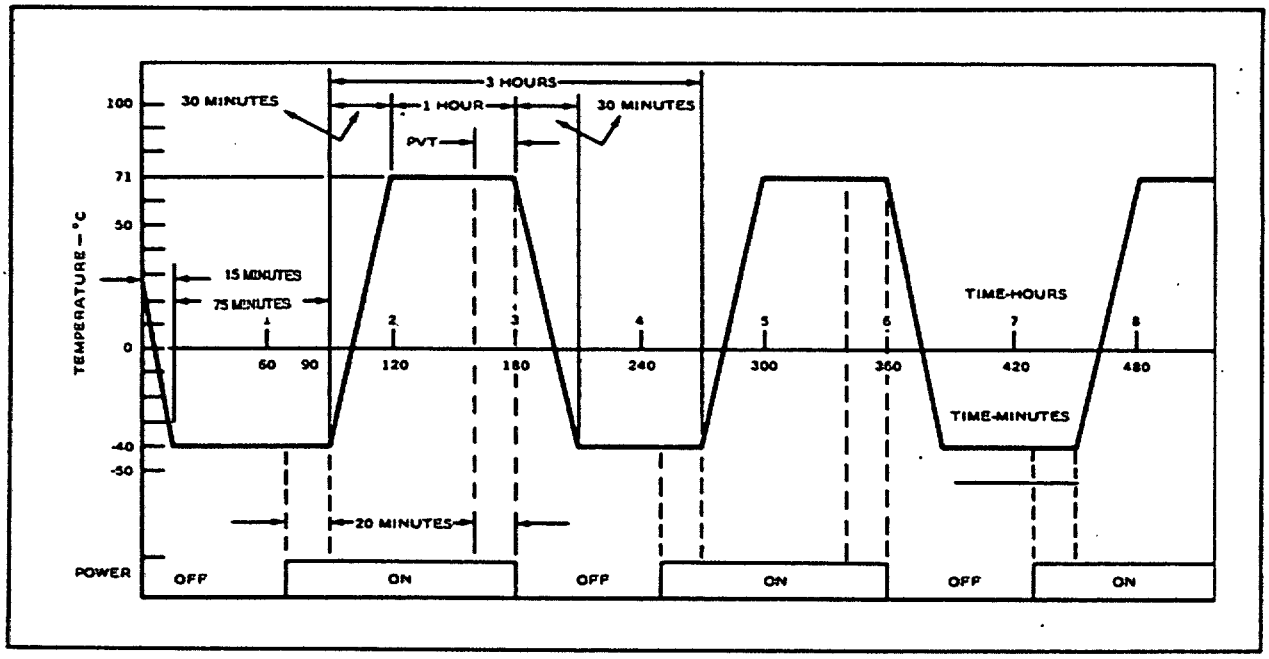


Figure 4. Temperature Cycling Curve.

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4.2.4 Performance measurements. Before, during and after the environmental inspection, where called for in an approved test procedure, the following tests shall be performed. Performance results and recorded data specifically required under the individual test shall be provided.

Test	Paragraph
Primary Input Power	4.2.3.1
Loudspeaker Amplifier	4.2.3.4
Squelch	4.2.3.5
Compatibility	4.2.3.8

The Government reserves the right to perform any non-environmental test listed in Table VIII at any point in the sequence of the environmental inspections. Failure to satisfy any requirement shall be a basis for rejection.

4.2.4.1 Noncompliance. If a sample unit fails the tests specified in the sample test column of Table VIII, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and the details of the corrective action taken to correct units of production which were manufactured with the same materials and processes under the same conditions. If the Government inspector does not consider that the corrective action will enable the product to meet the specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the Contracting Officer. Approval to ship the lot may be withheld at the discretion of the Government pending the decision from the Contracting Officer on the adequacy of corrective action. The Government reserves the right to require selection of new samples and a new sample test to be performed per Table VIII after completion of any reworking of the lot.

4.2.4.2 Corrective action. Corrective action found necessary by the contractor or command or agency concerned as a result of test failure shall be at the contractor's expense. Further, the contractor shall retrofit the corrective action into the production lot from which the failed item was originally selected. Applicability of the corrective action and retrofit to previous production shall be determined by the command or agency concerned. Cost of all retrofits and subsequent reinspections shall be born by the contractor.

4.2.5 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

~~a. First Article Inspection of Packaging.~~
b. Quality Conformance Inspection of Packaging

4.2.5.1 First article inspection of packaging. Unless otherwise specified in the contract, First Article Inspection of Packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

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4.2.5.2 Quality conformance inspection of packaging.

4.2.5.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.2.5.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified herein.

4.2.5.2.3 Packaging inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in Table III, "PACKING INSPECTION PROVISIONS." Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and Acceptable Quality level of 4.0 percent defective.

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-2073-2, coded as follows:

5.1.1 Amplifier-power Supply AM-7152/GRC-213.

5.1.1.1 Level A. 10-1-00-00-NS-X-ED.

5.1.1.2 Level B. 10-1-00-00-NS-X-ED.

5.1.2 Antenna Switching Unit SA-2365/GRC-213.

5.1.2.1 Level A. 10-1-00-00-NS-X-ED.

5.1.2.2 Level B. 10-1-00-00-NS-X-ED.

5.1.3 Vehicular Mount MT-6234/GRC-213.

5.1.3.1 Level A. 10-1-00-00-NS-X-ED.

5.1.3.2 Level B. 10-1-00-00-NS-X-ED.

5.1.4 Cable assemblies. Unit pack each in accordance with MIL-C-55442.

5.1.5 Consolidation. Place the items, unit packed as specified, within a close-fitting box conforming to PPP-B-636, W5c. Closure shall be in accordance with the appendix of the box specification.

5.2 Packaging and marking. Packing and marking shall be in accordance with MIL-E-55585.

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6. NOTES

6.1 Intended use. The radio set and vehicular adapter kit are intended for use by military forces to establish and maintain communications with their headquarters, adjacent units, and other conventional forces equipped with tactical HF single-sideband radio sets.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirement.

a. Title, number, and date of this specification.

b. Color of unit.

c. Materials, processes, and parts (see 3.3.2.10).

d. Standard conditions (see 3.3.2.7.3).

e. Number of first article samples to be submitted if other than specified in 4.1.2.1.3.

f. Reinspection of conforming environmental sample units (see 4.1.2.1.5.2).

g. Level A or level B preservation and packaging (see Section 5).

h. When the rough handling test is required (see 4.2.5).

i. When first article packaging inspection reports require Acquisition Activity approval prior to production unit packing.

6.2.2 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage, and handling.

6.2.3 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.2.4 Contract data requirements. Data generated by this document are not deliverable unless specified in the Contract Data Requirements List (DD Form 1423) or the contract schedule. The data required by this specification include, but are not restricted to, the following:

a. Maintainability demonstration report (see 3.3.2.6).
b. Quality program requirement (see 4.1.1.1).
c. Quality plan (see 4.1.1.2).
d. Test procedures (see 4.1.1.4).
e. Test plans (see 4.1.2.1.1).
f. First article inspection report (see 4.1.2.1.3.1).

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6.3 First article. Invitations for bids shall provide that the Government reserves the right to waive the requirement for first article samples to those bidders offering a product which has been previously procured or owned by the Government, and that bidders offering such products who wish prior Government approval to demonstrate that such products are presently appropriate for the pending procurement.

6.4 Subject term (key word) listing.

Distribution Statement A
Amplifier, Power Supply
Amplifier, Power Supply AM-7152/GRC-213
Adapter Kit, Vehicle
Adapter Kit, Vehicle MK-2154/GRC-213

Custodian

Army-CR

Preparing Activity:

Army-CR

(Project 5820-A753)

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TABLE III. Interface definition (See Paragraph 3.1.2.1).

Connector J2	Line Function	Line Source	Line Conditions	Function
A	MIC Shield	J102A		
B	RCVR Audio In	RT-1209	During receive audio output signal is fed through the audio filter to the amp.	2mW minimum across 500 ohm ± 20% frequency response: 500 to 3000 Hz Volume Control: -40 dB relative to max. output
C	PTT	J102S	During transmit GND through diode and EMI filter.	
D	Audio TX	J102U	During Transmit, MIC audio signal from VIC-1 is fed through attenuator and audio filter to Rcvr/Excr. RT-1209	50 dB attenuation (input signal -5 dBm; output signal -5 dBm) across 150 ohms into RT-1209
E	CW KEY			
F	KY-65 PTT	AM-6879(LP)	to KY-65 PTT applied	Grounded when PTT

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TABLE IV. Interface definition (See Paragraph 3.1.2.2).

Connector J104	Line Function	Line Source	Line Conditions	Function
A	GND	AM-7152		
B	GND	AM-7152		
C	GND	AM-7152		
D	Spare			
F	Radio Power	AM-7152	+26.5 Vdc nom. to PRC-104 when power switch is "ON"	5.5 amps Max. 20 Vdc Minimum 32 Vdc Maximum
H	Radio Power	AM-7152		
J	Radio Power	AM-7152		
F	Spare			
K	Spare			

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TABLE V. Interface definition (See Paragraph 3.1.2.5).

Connector J102	Line Function	Line Source	Line Conditions	Function
A	GND	AM-1780 J503	GND in AM-1780	
B	Unprotected primary output power positive battery voltage	J103	Provides power to AM-1780 and VIC-1 Intercomm. Sys. when cable XC-4723 is connected to J501 in AM-1780	
H	Unmuted audio	Audio Sq. Amp.	Side tone Rcvr. audio output	100mW nominal into 600 ohms
K	Monitor Amp. output	Audio Sq. Amp.	Side tone Rcvr, audio output	0.24 Vrms nominal into 150 ohms.
S	PTT line	VIC-1 Control Box	Provides GND to RT-1209/ PRC-104 and audio and squelch amp. when in transmit through EMI filter and a diode.	
T	Retransmit	Audio and Sq. Amp.	Provides GND to VIC-1(V) Intercomm. sys. when received HF signal on PRC-104 breaks squelch	Ground through EMI filter
U	Audio TX	VIC-1 Control Box	Provides audio signal from MIC to RT-1209/PRC-104	0.22 Vrms across 150 ohms.
Pin C connected to Pin D				Required for proper operation with VIC-1 system

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TABLE VI. Interface definition (See Paragraph 3.1.2.6).

Connector J1	Line Function	Line Source	Line Conditions	Function
A	Speaker	Trans.	Speaker return	
B	Sidetone audio output	Audio squelch card		100 mW audio output power.
C	PTT	J102S	Parallel to J2-C	During transmit GND through diode and EMI filter
D	Audio TX	J102U	Parallel to J2-D	
E	1 watt audio output	Sq. Amp.	1 watt squelched audio to loudspeaker	

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TABLE VII. AN/PRC-104 compatible vehicles and communication systems.

Vehicle	Communication System						
	VRC-43	VRC-44	VRC-46	VRC-53	VRC-64	GRC-125	GRC-160
JEEP M151			x	x	x	x	x
TRUCK M561			x	x	x	x	x
APC M113	x			x	x	x	x
APC M577	x		x	x	x	x	x

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TABLE VIII. Test.

Test	Requirement Paragraph	Test Paragraph	First Article	Prod	Sample	Environ
* Mechanical Inspection	3.3.2	4.1.2.1.4.3	x		x	
** Input Power	3.2.1.1.1	4.2.3.1	x	x		
** Input Voltage	3.2.1	4.2.3.3	x		x	
* Reversed Polarity	3.2.1.1.2	4.2.3.2	x			
** Protected Power Output	3.2.1.1.3		x			
** Max. Protected Output Current	3.2.1.1.6		x			
* Overload Protection	3.2.1.1.7		x			
** Auxiliary Power Output	3.2.1.1.8		x	x		
** Audio Amp. Input	3.2.1.2.4		x		x	
** Covers and Seals	3.3.4		x		x	
** Audio Amp. Output	3.2.1.2.5		x		x	
** AF Reduction (Muting)	3.2.1.5.1		x		x	
** Squelch Sensitivity	3.2.1.5.2	4.2.3.5	x		x	

* System
** AM/7152/GRC-213 (Amplifier Power Supply)
.* AM-7152/GRC-213, SA 2365/GRC-213, MT-6234/GRC-213
.. AM-7152/GRC-213, SA 2365/GRC-213

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TABLE VIII. Test - Continued.

Test	Requirement Paragraph	Test Paragraph	First Article	Prod	Sample	Environ
** Squelch Hysteresis	3.2.1.5.3		x			
** Sq. Attack and Release Times	3.2.1.5.4		x		x	
** Audio Amp. Freq. Resp.	3.2.1.2.6		x			
** Speaker ON/OFF	3.2.1.2.2		x	x		
** Headphone Output	3.2.1.4.1		x	x		
** Monitor Output (unmuted)	3.2.1.3.1		x	x		
** Headphone Output Dist.	3.2.1.4.1		x		x	
** Monitor Output Dist.	3.2.1.3.1		x		x	
** Headphone and Monitor Volume	3.2.1.2.3		x	x		
** Headphone Output Impedance	3.2.1.4.3		x			
** Monitor Output Impedance	3.2.1.3.3		x			
* Retransmit	3.2.1.7		x	x		
** Muting	3.2.1.8		x	x		
* PTT Control	3.2.1.9		x	x		
** Transmitter Audio Output	3.2.1.6		x	x		

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TABLE VIII. Test - Continued.

Test	Requirement Paragraph	Test Paragraph	First Article	Prod	Sample	Environ
* EMC	3.2.1.10	4.2.3.6	x			
* Input Voltage	3.2.2.1		x			
** Output Power	3.2.2.2		x			
** Hum & Ripple	3.2.2.3		x			
* Receiver Sensitivity	3.2.2.4		x			
* Reliability	3.3.2.5	4.2.3.13	x			
* Maintain-ability	3.3.2.6	4.2.3.7	x			
* Compati-ability	3.3.2.7.2	4.2.3.8	x			
* Ambient Temperature (Operating)	3.3.2.7.4.1	4.2.2.1	x			x
* Non-Operating	3.3.2.7.5(a)	4.2.2.1	x			x
* Humidity	3.3.2.7.4.3	4.2.2.2	x			x
* Altitude Operating	3.3.2.7.4.4	4.2.2.4	x			x
* Non-Operating	3.3.2.7.5(b)		x			x
* Vibration	3.3.2.7.4.5	4.2.2.5	x			x
* Shock Bench Handling	3.3.2.7.4.6	4.2.2.6	x			x
* Shock Drop	3.3.2.7.4.7	4.2.2.7	x			x
* Shock	3.3.2.7.4.8	4.2.2.8	x			x
* Bounce	3.3.2.7.4.8(a)	4.2.2.8.1	x			x

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TABLE VIII. Test - Continued.

Test	Requirement Paragraph	Test Paragraph	First Article	Prod	Sample	Environ
* Shock Ballistic	3.3.2.7.4.8(b)	4.2.2.8.2	x			x
* Dust	3.3.2.7.4.9	4.2.2.9	x			
.. Leakage, Immersion	3.3.2.7.4.10	4.2.2.10	x			x
* Fungus	3.3.2.7.4.11	4.2.2.11	x			
* Salt Fog	3.3.2.7.4.12	4.2.2.12	x			x
** Control Life	3.3.2.7.6	4.2.3.9	x			
** Control End Stops	3.3.2.7.6.1	4.2.3.10	x			
** Burn-In	3.3.2.13	4.2.3.11		x		
* Interchange-ability	3.3.6	4.2.3.12			x	
Workmanship Screen		4.2.3.14		x		
Temperature Cycling		4.2.3.15		x		

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TABLE IX. Packaging inspection provisions.

No.	Characteristic	Method of Inspection
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & Measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-49273	Amplifier, Power Supply AM-7152/GRC-213 & Adapter Kit, Vehicle

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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